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Development of a Blended Wing Body Aircraft with Hydrogen-Electric Hybrid Distributed Propulsion

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This paper introduces a 112-passenger blended wing body aircraft concept as part of the Integrated Zero Emission Aviation academic-industry collaboration to achieve NASA's goals of net zero emissions by the year 2050. This effort considers a novel liquid hydrogen-electric hybrid distributed system that generates propulsion power via a combination of liquid hydrogen-burning turboelectric generators and hydrogen fuel cells. We present the concept architecture and modeling assumptions in this paper and discuss our initial aerodynamic optimization of the clean airframe outer mold line using gradient-free surrogate-based optimization. We include and discuss the aerodynamic optimization results, system-level performance characteristics, and weight breakdown of the aircraft. The proposed concept has a roughly 34% reduction in energy burn compared to a notional model of the current state-of-the-art E190-E2 regional jet. The work described in this paper serves as a foundation for future design refinements and related system/sub-system level studies as part of this multi-year initiative.

Nomenclature

C_D = drag coefficient

C_L = lift coefficient

C_M = pitching moment coefficient

d = number of design variables

k = dimension of the active variable vector

L/D = lift over drag

M_∞ = freestream flow Mach number

n = number of samples

$\mathbf{W}$ = matrix representing the active subspace

$\mathbf{x}$ = vector representing a single design point

$\mathbf{z}$ = vector representing the active variables

Subscripts

cg = center of gravity

r = outboard wing root

t = outboard wing tip

w = outboard wing

wl = winglet

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I. Introduction

Global CO_2 emissions from aviation have increased dramatically despite vast improvements to gas turbine engines [1]. Aviation produced 2.4% of global CO_2 emissions in 2018, with the U.S. accounting for nearly one-fourth of the total. Aviation is predicted to produce 25% of global CO_2 emissions by 2050 [2]. In addition, the use of air in high-temperature/high-pressure combustion produces NOx, which reduces air quality and is also a greenhouse gas. NOx emissions are more difficult to reduce than other pollutants [3]. Increasing global emissions from the aviation industry has led to a push towards stricter emissions laws. The goal is to achieve net-zero greenhouse gas emissions from the U.S. aviation sector by 2050. The FAA has several approaches for how to reach this goal. These include new aircraft and engine technologies, increasing production of sustainable aviation fuel, increasing operations efficiency, moving towards electric propulsion, and improving fuel efficiency at airports [4].

Hydrogen as propulsion fuel is a very attractive option with zero CO_2 emissions, a higher specific energy than batteries, and lower NOx emissions than ammonia. Liquid hydrogen (LH_2) is preferable to gaseous hydrogen due to its higher volumetric energy density. To convert LH_2 fuel into useful energy, there are three primary options. The first involves combustion in a modified turbofan engine (LH_2 to shaft power), which, although very efficient in large gas turbine engines, still produces NOx emissions and contrails [5] thereby reducing hydrogen's otherwise favorable impact on climate change and air quality.

The second option is hydrogen fuel cells which are efficient and avoid NOx and contrails. Entities such as CHEETA [6] and Airbus [7], are exploring fuel cell–powered aircraft. Fuel cells are also particularly advantageous for meeting non-propulsion-related electrical power requirements such as for the cabin and avionics. In current aircraft, generators in the gas turbine engines and auxiliary power units (APUs) provide electric power. The size of these generators is proportional to the electric power requirements. However, engine size limitations will restrict the growth of these generators in future aircraft [8]. As such, fuel cells can be used to replace these generators and APUs as a source of non-propulsion electric power on future aircraft.

The use of fuel cells for aeronautical applications has challenges that include weight and balance, designing a thermal management system, managing power flows with multiple power sources, and maintaining safe operations [9]. In 2008, Boeing successfully completed flight tests of the first manned hydrogen fuel cell airplane. They attempted to address the aforementioned challenges and prove the feasibility of integrated fuel cells for aviation applications. These tests showed that the biggest barriers that will prevent the implementation of fuel cells in aerospace systems in the future are related to hydrogen storage and component integration within a restricted volume [9].

The biggest challenges with LH_2 storage stem from its 20 Kelvin temperature requirement to maintain its liquid state and its substantially lower volumetric energy density than Jet-A. These characteristics result in a large fuel tank volume requirement, part of which is lost for tank insulation. A key enabler for LH_2 storage is a blended wing body (BWB) airframe, which offers significantly more volume flexibility than a tube and wing aircraft. Coupled with a higher lift-to-drag ratio (L/D) compared to the tube and wing [10], the BWB also enables more efficient use of LH_2, allowing the aircraft to fly further for a fixed volume of fuel.

The third option for converting LH_2 into propulsive energy involves combustion in a modified turbofan engine coupled with electric generators (turboelectric). The propulsive efficiency of the turboelectric concept can be improved further through distributed propulsion systems. NASA's N3-X BWB aircraft concept used two Jet-A burning turbogenerators mounted on the wingtips to generate electricity that powered an array of boundary layer ingesting superconducting motor-driven fans spanning the upper trailing edge of the centerbody section [11]. The multi-fan approach allowed for a very high effective bypass ratio while retaining the efficiency of large core engines. The multi-fan array provided flexibility in terms of positioning the generators and propulsors, low fan noise, and reduced core jet noise [12]. However, some disadvantages to this approach include the weight of electric components and its reliance on progress in the field of electric device technology [13].

A combination of fuel cells and distributed electric propulsion can also be considered as means to improve fuel efficiency and reduce emissions. An aircraft concept that combines the BWB with fuel cells and distributed propulsion is NASA's Quiet Green Transport Concept B [14]. This concept uses the BWB airframe as an aerodynamically efficient configuration that also provides noise shielding with top-mounted engines. The integration of highly reliable and maintainable long-life LH_2 fuel systems into the airframe was, however, identified as the main hurdle in the viability of this concept in the near term future [14]. The CHEETA concept introduced above is another example. It is a 180 pax hydrogen aircraft designed by the University of Illinois and several other partner universities through a university leadership initiative (ULI) funded by NASA [6]. This concept employs electric distributed propulsion, hydrogen fuel with tanks distributed throughout a novel airframe layout, and is powered by solid oxide fuel cells (SOFC). The propulsion system utilizes novel cryogenic electric technologies due to the high power and stringent weight requirements

for the electric components. The primary challenges also appear to be the weights of the fuel cell system, driven by fuel cell technology, and thermal challenges due to the high power requirements and associated waste heat of the fuel cells.

In this paper, we introduce the Integrated Zero Emission Aviation (IZEA) concept, which embraces a flexible hybrid power generation architecture to address some of the challenges associated with hydrogen aircraft and fuel cell usage. The core idea is to utilize both turbine engines and fuel cells to balance the overall specific power (weight) of the system and the emissions produced. The concept leverages other technologies previously discussed: a blended wing body architecture; electric distributed propulsion with boundary layer ingestion; turboelectric power generation for high power and low weight; hydrogen fuel cells for low NOx and on-board power generation; and a multi-temperature superconducting power system. The IZEA initiative is being led by Florida State University and involves other partner universities in the design of the power system and aircraft components. The project is a 5-year effort.

The purpose of this paper is to introduce the core requirements for the IZEA aircraft concept and an initial aircraft design based on the benchmark project component goals. In Sec. II we present an overview of the IZEA concept architecture and our assumptions and modeling approach for the various components. We elaborate further on the aerodynamic design and analysis method in Sec. III. An initial airframe optimization is conducted, the results of which are presented along with the resultant system-level performance characteristics discussed in Sec. IV. Note that in the context of the overall IZEA program, these results are preliminary and the concept will be refined further in future work as part of this multi-year initiative. As such, this work lays out the foundation for future development efforts, and we conclude with some remarks on these future plans for the IZEA concept in Sec.V.

II. IZEA Concept

A. Aircraft Requirements and Mission

The IZEA aircraft is a regional jet concept that is similar in capability to the Embraer E190-E2 airplane. The target design range requirement for this configuration with its passenger capacity of 112 (2 class layout) is 2000 nmi, though the exact range will fall out of the systems analysis. Figure 1 presents the design and reserve mission profiles for the IZEA concept. This aircraft cruise climbs at a specified cruise Mach, with an altitude range roughly between 41,000 ft. and 42,000 ft. The top of climb/start of cruise altitude is a fallout of the engine thrust capability, aerodynamic performance of the airframe, maximum takeoff weight, and design range. The design mission range is driven by the efficiency of the vehicle, given a fixed available usable fuel capacity. Note that the available fuel volume is used for a combined full-design mission and reserve mission. The reserve mission has a 200 nmi range to an alternate airport,

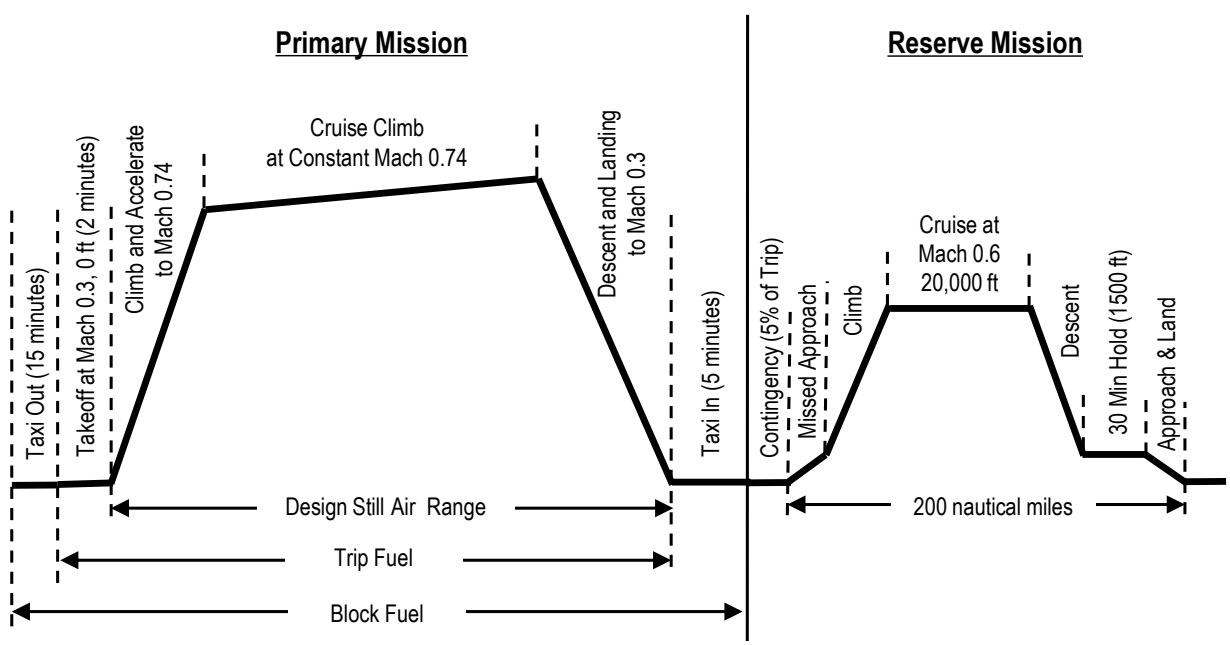


Fig. 1 Notional IZEA mission profile

cruising at Mach 0.6 at 20,000 ft. The reserve mission also includes a 30-minute hold at 1,500 ft. The total reserve fuel is the value required to fly the mission plus an additional 5% of the trip fuel. We will also define and evaluate a 900 nmi mission as being representative of a typical economic mission flown.

B. Configuration Design

The IZEA concept vehicle with the bounding box representations of its internal components is shown in Fig. 2. The aircraft's length is 93.5 ft with a wingspan of 158 ft. The wing folds at roughly 75% wingspan to be within the 118ft ICAO 3C gate compatibility requirements [15]. The passenger cabin is a single-deck design, which is shown to be a better option for low passenger/medium-long range as compared to a conventional double-deck cabin that tends to fare better for high passenger short-medium ranges [16]. The landing gear in the stowed position will be located at the main passenger cabin level, made possible by the use of technology like the pivot-piston landing gear proposed for the ASCENT-1000 BWB [17]. The passenger cabin has a two-class passenger layout with a total of 112 passengers, 100 in economy in 10 abreast arrangement and 12 passengers in the business class section in 4 abreast arrangement. On each side of the aircraft, one type-A main exit is at the front and one type-I emergency exit is placed at the rear in compliance with 14 CFR § 25.807 [18]. One cylindrical LH_2 tank and one bay of three LD-3 cargo containers are placed on each side of the cabin. Two turbogenerators contain a turboshaft engine driving a superconducting generator for electricity. The electric power from the turbogenerators is distributed via superconducting cables to an array of superconducting motor-driven propulsor fans spanning the entire center wing-body width near the trailing edge. The placement of turbogenerators next to the fan array on the upper side of the airframe, as shown in Fig. 2, as opposed to on the wing tips or further up on the centerbody, simplifies the cables and cooling hardware layout, resulting in some weight savings. The exact fan array and turbogenerator location, fan array aerodynamic design, and boundary layer ingestion effects will be the subject of future modeling work in this multi-year effort.

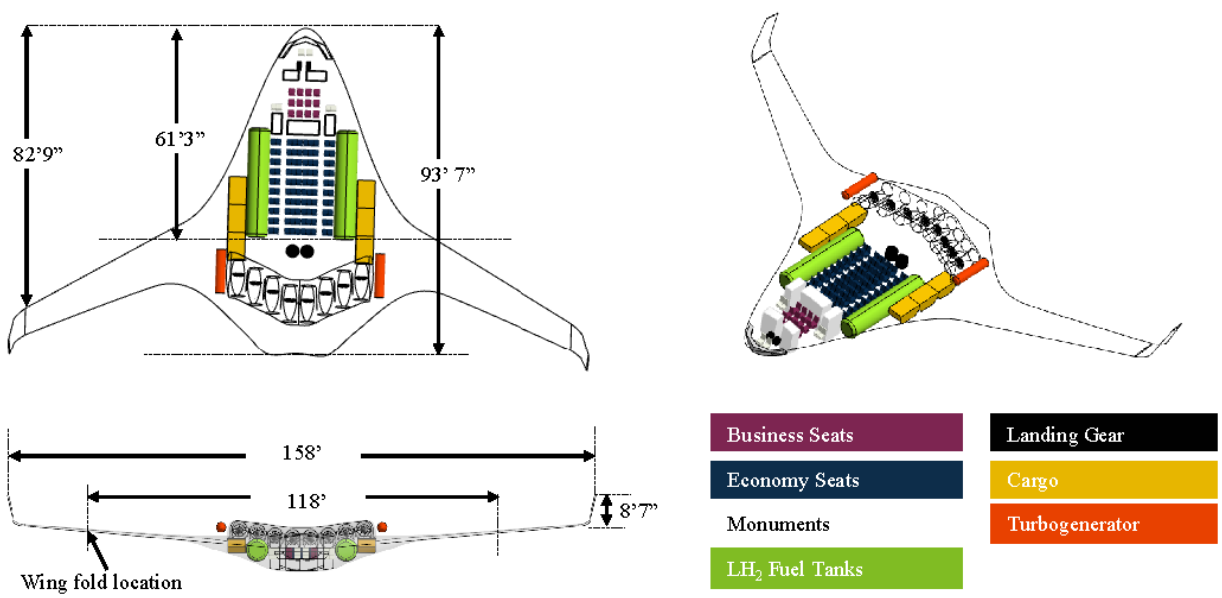


Fig. 2 IZEA aircraft configuration (with a notional electric fan array)

C. Propulsion System

1. Architecture Definition and Hybridization

A detailed propulsion system architecture is in development for the aircraft, including thermal flows, hydrogen flows, and electrical systems. The primary architecture, however, has the following major components: hydrogen-burning turboelectric engine and associated high-temperature super-conducting (HTS) generator (one left-hand side and one right-hand side); proton-exchange membrane fuel cell utilizing hydrogen and associated thermal management system;

HTS power distribution network and protection devices; thermal management system (heat exchangers, pumps, etc); fan drive, motor, and associated thermal system; eight propulsion fans (rotor blades, disks, bearings, ducting, nacelles, etc.); and hydrogen fuel system (tanks, plumbing, accessories, etc).

The core design feature of the hybrid architecture is the degree of hybridization of power between the turboelectric generator (combustion) and fuel cell. We parameterize this by defining a hybridization split variable at the top-of-climb point between engine power output and fuel cell power output. This variable has a maximum of 100% which would correspond to a fully electric cruise, wherein the fuel cell would be entirely sized by the cruise power of the airplane. In theory, the hybridization can be lowered to zero corresponding to the purely turboelectric case. However, a minimum value of 17% is chosen in order to maintain a fuel cell of sufficient size to taxi and descend the aircraft without an assist from the engine to minimize overall NOx production. We call these two bounding cases the IZEA "Light" (17% cruise hybridization) and IZEA "Pure" (100% cruise hybridization). In between these two cases, one can define a continuous distribution of potential concepts with differing degrees of hybridization. The project is interested in exploring this design space to best balance energy use and NOx production to minimize overall global warming impact. We call this design the IZEA "Balanced" design. This paper will discuss the design of the IZEA Light aircraft as much more work has been conducted on this concept than the IZEA Pure and the fuel cell requirements are significantly less challenging for the former.

2. *Sizing Requirements*

For the propulsion system, it is first necessary to define the thrust requirements for the system. These thrust requirements, together with a fan design, will translate to power, torque, and RPM requirements for the electric machines and other electrical and power generation components. For thrust, the system must be designed to accommodate the following requirements: minimum balanced field length of 6300 ft to maintain similarity with other competitive regional jet performance; minimum FAA climb gradient for an equivalent engine inoperative condition (since the system is distributed) per CFR § 25.121; and minimum initial cruise altitude capability of 300 ft/min at the aircraft design cruise altitude.

We calculate the balanced field length (BFL) from equation 5-91 in Torenbeek [19] using parameters from the aircraft. We calculate the second segment climb gradient from the following equation:

$$\frac{T_2}{W} = \gamma + \frac{C_D}{C_L} + \Delta\frac{C_D}{C_D}_{\text{OEI}} \tag{1}$$

where C_D/C_L is determined at the low-speed condition and includes the difference in drag due to four inoperative fans (worst case condition). The required thrust-to-weight ratio to satisfy the BFL constraint and the second segment climb gradient is then chosen. In general, the BFL constraint was much more stringent than the second segment climb requirement, due to the fact that there is significantly less drag at high lift for the blended wing body at point 2 in the flight profile. Figure 3 shows a point performance assessment of the airplane with various constraints applied while

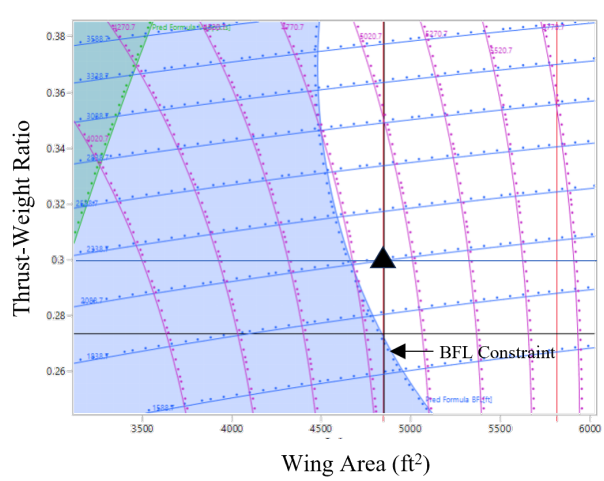


Fig. 3 Preliminary point performance constraint results for the IZEA Light aircraft

varying wing area and thrust-to-weight ratio. The BFL constraint dictates the thrust-to-weight ratio selected for the concept at 27.3%, however, we selected a more conservative value of 30.3% for initial sizing to allow for potential weight growth. This corresponds to a reference sea-level static thrust for the aircraft of 36,290 lbf for the IZEA Light.

3. Fan Modeling

We modeled the fan system using the NPSS program [20]. The core model that was utilized is described in Gladin et al. [21]. For design point performance, the fan was sized at the top of climb point for thrust required. The parameters used for the fan design are shown in Table 1 for reference. We quantified off-design performance through the use of a static fan map, which was generated using the NASA CMPGEN program but was designed to be similar to that of the NASA SDT Fan [22]. We assume that there is a variable area nozzle in the fan array exhaust stream, which was also assumed for the NASA N3-X [11]. This is necessary for the purpose of maintaining proper fan stall margin and efficient performance at low altitude. Results of the fan model sizing and off-design performance show that the required motor output powers are 16.6 MW for takeoff and 8.13 MW for top of climb. The sized corrected airflow at the top of climb point was 44.55 lb/s. Assuming a specific corrected airflow of 40 lbm/s-ft^2 and a hub-to-tip ratio of 0.3, then the fan diameter is 3.77 ft.

Table 1 Fan design parameters for the IZEA aircraft

Parameter	Value
Fan pressure ratio	1.3
Fan adiabatic efficiency	0.929
Fan inlet recovery	0.998
Fan exhaust duct pressure loss	0.005
Fan nozzle velocity coefficient	0.9954
Fan nozzle discharge coefficient	0.9588

4. Electrical Components

In the initial sizing and requirements definition phase, the aircraft is assumed to have groupings of electrical components and associated key performance parameters shown in Table 2. Each grouping is assumed to include uninstalled performance of the components but to include all required components for the function of each grouping. We sized the components according to their initial rated power and the associated specific power. The efficiencies shown in Table 2 are used when determining power flows, thermal management heat loads, and ultimately fuel flows.

Table 2 Electrical component initial target key performance parameters

Components	Target Specific Power (kW/kg)	Efficiencies
Fuel cell stack	3	0.55
Turboelectric generator	11	0.6 (cruise, lower at part power)
Motor + thermal management	8	-
Drive	30	0.985
HTS distribution and protection	25	0.999
Fuel cell cooling	5.5	-

A more detailed analysis of the specific technologies chosen for each grouping will be performed by University partners at Florida State University, the University of Buffalo, and the University of Kentucky. Later results and detailed models of these component studies will be integrated into the power system and weight model and compared against this initial assessment.

5. Power Generation Modeling

The primary power systems for the IZEA concept are two hydrogen-powered turboelectric generators and a fuel cell. These systems are modeled using the GT-HEAT tool as well within NPSS. For the hydrogen properties, we used the JANAF thermodynamic package, which we found to be more appropriate for modeling hydrogen fuel properties. The assumed architecture of the engine is a three-spool configuration with two gas generator spools and one free power turbine spool that drives a HTS generator. The generator is connected to a notional distribution network as shown in Fig. 4 which ties in a prescribed percentage of fuel cell power at the electrical bus to drive the fan motor. The architecture also passes the hydrogen fuel, which exits the superconducting thermal management system (TMS) heat exchangers at approximately 353 Kelvin. Waste heat from the exhaust of the thermal engine is exchanged into hydrogen fuel in a method similar to that used in the engine designs of Brewer [23]. Future work will look at considering cooled cooling air which is shown in the schematic but not employed in this study.

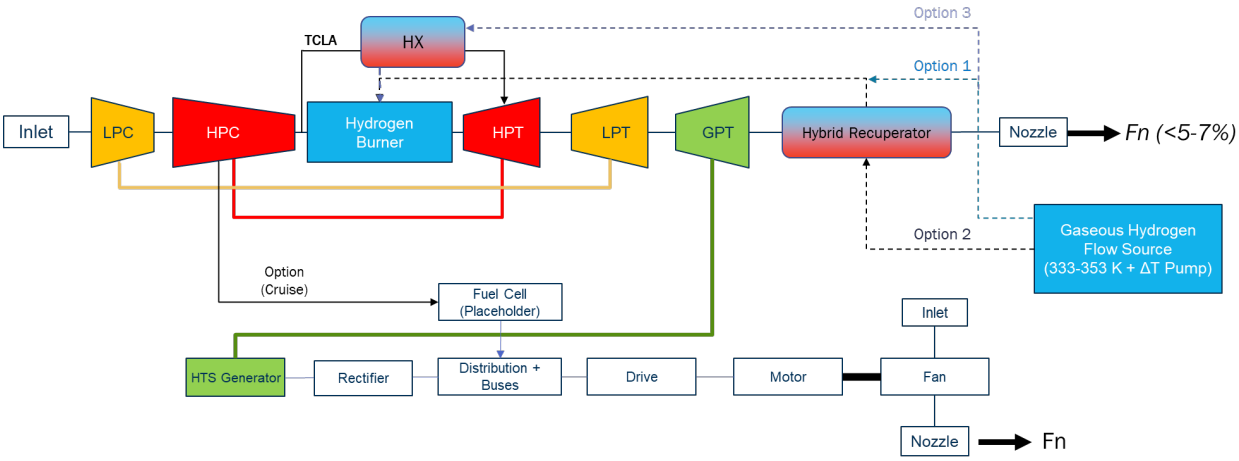


Fig. 4 Turboelectric engine modeling schematic

We modeled the thermal engine components using standard NPSS elements, with some custom inputs. For the compressors, we selected design point pressure ratios and efficiencies consistent with the size of the respective machines and a future technology assuming a 2050 entry into service (EIS). For the turbines, polytropic efficiencies at the design point were determined based on the size of the machines, and cooling flows were determined using the NASA CoolIt NPSS program. Ceramic Matrix Composites (CMCs) were applied to both rotors and stators of the high pressure turbine (HPT) because of the assumed 2050 EIS. For off-design, notional compressor maps with Reynolds effects for the engine were derived from the NASA CMPGEN program with appropriate inputs for the designs of the compressors. For the burner, a custom element was written to model the hydrogen burner, with assumptions as shown in Table 3.

We modeled the electrical chain using simplified efficiency assumptions with appropriate power balance equations that transfer power from the HTS generator and fuel cells to the electric motor. The fuel cell is not explicitly modeled within the GT-HEAT tool as of yet, as that task is being conducted by the University of Buffalo, but we created a placeholder power model to allow for preliminary determination of hydrogen fuel flows assuming a mission average fuel cell efficiency of 55%. The fan model described previously is a separate component of the flow path and dictates power demand from the motor, which back propagates to power requirements for the HTS generator and fuel cell at any given point. Equations for continuity of energy and mass flow are solved by the native Newton-Rhapson solver within the NPSS program to ensure physical feasibility. A multi-design point procedure is employed to size the overall system for power at both takeoff condition and top-of-climb at the desired cruise altitude. The take-off condition generally sizes the power system ratings and the top-of-climb point sizes the fan and the fuel cell system which is defined as a fixed 17% of top-of-climb power.

D. LH_2 Tank Model

Our research partners at Florida State University are in the process of developing detailed LH_2 tank models and designs. In the meantime, we made some basic assumptions and set some requirements for the tanks to proceed with our outer mold line (OML) definition and system-level analysis. The aircraft contains two cylindrical LH_2 tanks with a

Table 3 Engine component parameters for the IZEA Light aircraft

	Parameter	SoA Value (Large Core)	SoA Value (@ Size)	Future Value (@ Size)
LPC	Pressure Ratio (-)	-	2.96	2.96
	Exit Corrected Flow (lb/s)	-	19.32	-
	Polytropic Efficiency (%)	92%	92%	93% (+1%)
HPC	Pressure Ratio	-	11.32	-
	Size (lb/s)	-	2.5	-
	Polytropic Efficiency (%)	92%	88.67% (-3.33%)	89.67% (+1%)
Brn	Delta Total Pressure	-	0.045	0.045
	Efficiency	-	0.9950	0.9950
HPT	Inlet Corrected Flow (lb/s)	-	3.83	3.83
	Polytropic Efficiency (%)	92%	89%	90.3%
	Cooling Flow (%CH/%NCH) *CMCs Assumed*	-	(3.2%/9%)	(2%/4.93%)
LPT/GPT	Inlet Corrected Flow (lb/s)	-	>10	>10
	Polytropic Efficiency (%)	92%	92%	93.25%
	Cooling Flow (%CH/%NCH)	-	LPT: (2%/3%) GPT: 0%	LPT: (1%/3%)

circular cross-section and 2:1 semi-ellipsoid head at each end. Each tank is 31.2 ft long and has a diameter of 6.3 ft. Our target gravimetric index for the tanks is 60%, with a required usable fuel weight of at least 6,390 lbs combined in both tanks. To maintain the low temperature necessary for liquid hydrogen storage, the tanks will be insulated to minimize thermal losses. Each tank and its associated plumbing is assumed to weigh about 2130 lbs.

E. System Level Modeling

We used FLOPS [24] to model the system-level performance of this aircraft. We integrated the propulsion system performance into FLOPS through an engine deck that contained estimates for fuel flow rate and net thrust at different Mach-altitude combinations for a sweep of engine throttle settings. The aerodynamic performance of the airframe was obtained through CFD in the form of C_L and C_D for multiple Mach-altitude combinations in the flight envelope. The drag polar generation process is described in Sec. III.D. We multiplied the drag estimates from CFD by a factor of 1.05 as a crude estimate for the profile and interference drag that one would get by integrating the fan array (which we did not model in CFD) on the airframe. In future work, we plan on including the fan array in the CFD model and optimizing it with powered boundary conditions. Regarding the mission shown in Fig. 1, the climb segment was optimized for minimum time to climb in FLOPS. The cruise climb segment for the design mission was optimized for a specific range, while the descent segment was run at maximum vehicle L/D. We used FLOPS internal weight equations for predicting the structural weight of the BWB as well as for most of the other non-structural weight components.

By default, FLOPS assumes a composite centerbody and a metal outboard wing and aft body. However, we assumed a pultruded rod, stitched, efficient, unitized structure (PRSEUS) centerbody, and a conventional composite outboard wing. The specified weight savings from PRSEUS and conventional composite technologies are depicted in Fig. 5. FLOPS wing and centerbody weight calibration factors, i.e., FRWI 1-4 and FRFU respectively are adjusted to account for the assumed impacts of conventional composites and PRSEUS.

The weight savings factors shown in Fig. 5 were derived from previous work associated with NASA Environmentally Responsible Aviation Integrated Technology Demonstrations (ERA ITDs), documented in [25]. As part of this work, estimates for the structural weight savings from PRSEUS relative to baseline orthogrid stiffened sandwich (BWB

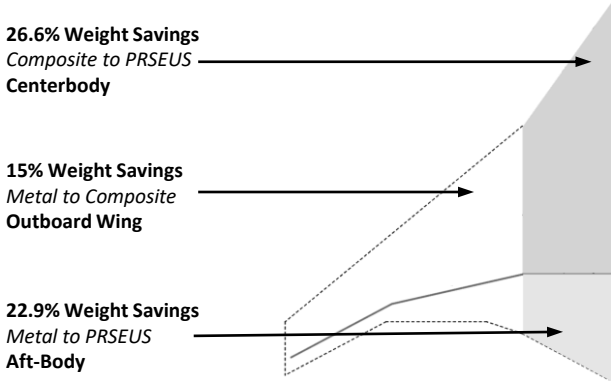


Fig. 5 Assumed weight savings going from FLOPS default assumptions to applied material technologies.

centerbody) and blade stiffened (BWB wing) composite configurations were generated. The weight factor values we used are representative of nominal values for the final milestone in the ERA ITD process. We assumed a weight savings of 15% when considering baseline composites over conventional metallic construction. Additionally, the weight reduction values of PRSEUS when compared to baseline composites are multiplicative with these factors. For example, the ERA ITD study shows a weight savings of 9.3% using PRSEUS instead of baseline composites for the aft body. Assuming the 15% savings from metal to baseline composite, the cumulative weight savings going from metal to PRSEUS is given by $1 - (1 - 0.093) \times (1 - 0.15) = 22.9\%$.

For propulsion system weights, each component weight was computed by first estimating the necessary maximum rated power from the critical peak power takeoff sizing point. We used the specific power targets from Table 2 to then estimate a target weight for each component. For TMS systems, the peak heat load was computed from the maximum rated power and efficiency of the components and then used to approximate the target weight allotment. To account for additional unforeseen components and any difficulties in meeting component target goals, a weight margin of 4857 lb was used to provide some buffer for additional propulsion weights. An additional 10% allotment was allowed for the installation of components and propulsive load-bearing members such as fan, engine, fuel cell mounting structure, and so on. The basic breakdown of propulsion weights for the IZEA Light follows in Table 4.

For systems weights, we used typical class 1 weight regression estimation equations from FLOPS to estimate the weights of aircraft electrics, hydraulics, pneumatics, ice protection, and environmental controls. The regression predicted weights were slightly modified and generally increased to match weight fractions from the Boeing SUGAR Ray [26], which is the closest conceptual aircraft to IZEA in class with published weight breakdowns. The auxiliary power unit weights were assumed to be zero since the fuel cell system can serve as a functional APU on the ground and in flight. However, fuel cell weights were bookkept in the propulsion category.

Table 4 IZEA Light propulsion system rated powers and weight estimation.

Component	Rated Pwr. (kW)	Tgt. kW/kg	Tgt. Mass (kg)	Margin	Installation	Total Wt. (kg)	Total Wt. (lbs)
Engine+Generator ×2	15528	11	1412	316	212	1939	4275
Fuel Cell	1400	3	467	104	70	641	1414
Fuel Cell Thermal	1146	5.5	208	47	31	286	631
Distribution	16911	25	676	151	102	929	2049
Motor Drive	16826	30	561	126	84	771	1699
Motor	16658	8	2082	466	312	2860	6306
Fans and Nacelles	16658	6.15	2709	606	406	3721	8203
						Propulsion Total (lbs)	24576

III. Aerodynamics Methodology

This section provides details on the aerodynamic modeling for the IZEA concept. The OML parameterization, optimization problem formulation and approach, mission drag polar generation, and CFD modeling details are presented in this section.

A. Geometry Parameterization

For the aerodynamic optimization, we consider an outer mold line (OML) that excludes the propulsion system and control surfaces. The parametric CAD model was developed using the open-source software Engineering Sketch Pad (ESP) [27]. We divided the OML into a total of 9 spanwise stations strategically placed to govern the curvature of the aircraft's leading edge, trailing edge, and winglet. The loft was generated by ESP's built-in *BLEND* function which generates a cubic spline surface passing through the airfoil stack. We chose four airfoil stations to govern the airfoil and planform definition during the optimization. Those stations are defined as Centerbody, Transition, Wing, and Winglet stations. The baseline airfoils at these stations are defined using eight class shape transformation (CST) coefficients [28] for each of the upper and lower surfaces. The airfoil sections for the stations in the Centerbody and Transition regions are linearly interpolated using the boundary stations. The outboard wing region maintains a constant airfoil section from the root to tip. During the optimization, the Centerbody station chord length is fixed. The chords of the remaining stations are obtained by running an optimizer with constraints on total planform area, wing/winglet taper-ratios, and Bézier control point locations. The optimization design space with all the parameters we varied is shown in Fig. 6.

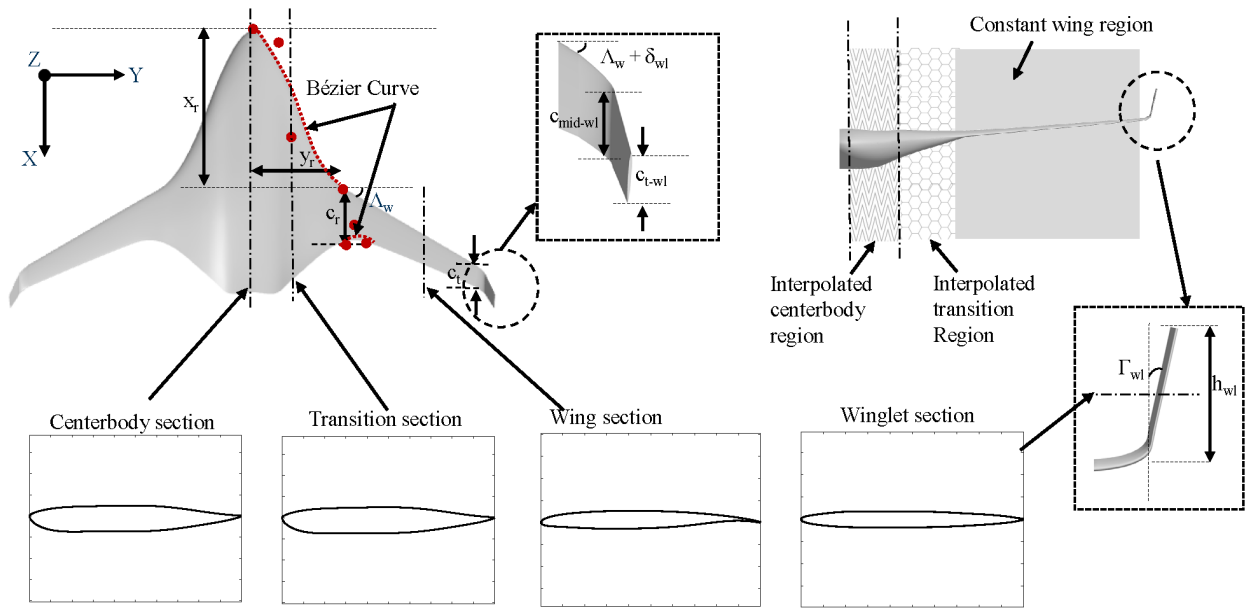


Fig. 6 Design optimization variables

B. Optimization Problem Formulation

For this optimization study, we minimized the following objective function:

$$-\frac{0.176}{C_D} + 10^6 {C_{M_{cg}}}^2 \tag{2}$$

evaluated at $M_\infty = 0.74$, 42,000 ft, where 0.176 was the initial estimate for the mid cruise lift coefficient. We enforced this lift target in the simulations. Initial component weight guesses enabled us to estimate the chordwise center of gravity location to be at 46.9 ft. Varying some component weight values with high uncertainty showed us that this center of gravity location is fairly insensitive to such changes and that the estimated location is adequate for the purposes of this initial effort. The reference area we used for the aerodynamic coefficients is half the planform area (given that we modeled half the geometry in CFD) and has a value of 2422 ft^2. The reference chord length we used for the pitching

moment calculations was set to 57.4 ft. Finally, we set a minimum static margin constraint of 5%. Additionally, for each design sample, the geometric constraints were verified inside ESP to ensure that the cabin, fuel tanks, and cargo bounding boxes were fully contained inside the OML.

We employed a gradient-free optimization strategy for the IZEA planform. Specifically, we used a modified variant of the efficient global optimization, or EGO, method [29, 30] to improve the aerodynamic performance of the airframe. The EGO method typically begins by evaluating warm start samples obtained from some space-filling design of experiments such as a Latin hypercube design. A Gaussian process (GP) regression model is then fit to those data. For each subsequent iteration, the expected improvement is maximized with respect to the design variables to yield the next point. This point is then evaluated and included in the training data to update the GP regression. In the modified version of the EGO approach that we adopted for this effort, i.e., the "Kriging believer" method [31], five new sample points were selected and evaluated before updating the GP model. This parallel-optimization approach minimizes calendar time compared to the traditional serial EGO. The first new sample point was selected by maximizing the expected improvement as in the original EGO method. The second point was also selected before this first point was evaluated. We temporarily treated the mean GP prediction at the first sample point as an observed data point and re-trained the GP regression. This updated GP model was then used to find the next sample point by maximizing the infill criterion. We repeated this process until all five desired sample points in a batch were selected and we then launched analyses simultaneously on the cluster. We stopped adaptive sampling when the expected improvement for the next sample point was below a certain percentage of the standard deviation in the objective function and when subsequent batches of cases failed to produce any meaningful change in expected improvement relative to the previous batch.

C. Dimensionality Reduction

To train a good GP-regressor, the necessary sampling density scales quickly with the input dimensionality. The training and evaluation of the GP model also becomes harder since the evaluation of the kernel matrix scales with $O(dn^2)$, where n is the number of samples and d is the number of design variables, and computing its inverse scales with $O(n^3)$. As such, increasing dimensionality results in expending significantly more resources to train and evaluate the regression model. Given that our problem had $d = 77$ design variables, we used the active subspace method to reduce the input dimensionality.

The active subspace method, initially proposed by [32], finds a linear subspace—or the active subspace—that best captures the dependence between an output of interest and the input space. From this subspace, the original regression problem can be reformulated into the following approximation

$$f(\mathbf{x}) \approx g(\mathbf{z}) = g(\mathbf{W}\mathbf{x}) \tag{3}$$

where $\mathbf{W}$ is a $d \times k$ basis representing the active subspace and $\mathbf{z}$ is a k-dimension vector containing the projection of the design variables $\mathbf{x}$ onto $\mathbf{W}$. The components of $\mathbf{z}$ are also called the active variables, and it is assumed that $k \ll d$. In the original work of Constantine et al., the basis $\mathbf{W}$ was computed using the gradient of the output variable sampled uniformly from the design space.

In this work, we leveraged gradient-free active subspace methods. Specifically, we used the manifold optimized active subspace defined by [33, 34]. This method finds the active subspace for a given problem by including the basis $\mathbf{W}$ in the trained parameters of a GP model. To reduce the cost of computing the active subspace, we extracted the basis $\mathbf{W}$ using samples from Euler CFD simulations, where a previous study demonstrated that the active subspace computed from inviscid results was a good approximation of the one computed from RANS results [35]. We used the Akaike information criterion to select the active subspace dimension.This resulted in an active subspace dimension of $k = 5$ instead of the initial dimension of $d = 77$. From the basis $\mathbf{W}$, we performed the adaptive sampling using RANS CFD as described in Sec. III.B on these active variables.

D. Multi-Fidelity Mission Drag Polar Generation

Given the optimized OML, we needed to incorporate the aerodynamic performance of this configuration for vehicle sizing and mission analysis. We needed a set of "mission drag polars" that provided estimates of C_D given C_L, Mach, and altitude for the entire operating envelope of the aircraft. While low fidelity semi-empirical methods, such as those described in Ref. [36], are appropriate for conventional tube and wing aircraft sizing and mission analysis, for more unconventional configurations such as the BWB, higher order physics modeling is needed. However, instead of generating the polars for each Mach-altitude combination using only RANS CFD, we leveraged multi-fidelity approaches to alleviate most of the computational burden with cheaper Euler simulations and interspersed the dataset with a handful

of RANS cases. Through hierarchical Kriging [37], the low-fidelity and high-fidelity datasets were combined into one drag polar table for the optimized configuration that provided the necessary aerodynamic information for the mission analysis code. Details of the multi-fidelity polar generation methodology can be found in Ref. [38].

E. CFD Modeling

Given our multi-fidelity aerodynamic design and analysis approach, we needed to develop CFD templates for both the viscous and inviscid runs. We used Siemens STAR-CCM+* for the Reynolds-averaged Navier Stokes (RANS) CFD simulations, assuming steady-state and standard atmosphere conditions. We chose an implicit time integration scheme with a third-order Monotonic Upstream-centered Scheme for Conservation Laws (MUSCL) central difference spatial discretization, with the advection upstream splitting method, flux vector splitting (AUSM+ FVS) [39] and the Venkatakrishnan limiter [40] for evaluating the inviscid fluxes. We assumed fully turbulent flow and selected the k-ω shear stress transport (SST) turbulence model [41]. We set up a hemispherical freestream boundary with a radius that was roughly 70 times the centerbody root chord length. The CFL number was automatically controlled by STAR-CCM+ for optimal convergence. We defined the stopping conditions as follows: the change in C_D is less than one drag count

*https://plm.sw.siemens.com/en-US/simcenter/fluids-thermal-simulation/star-ccm/

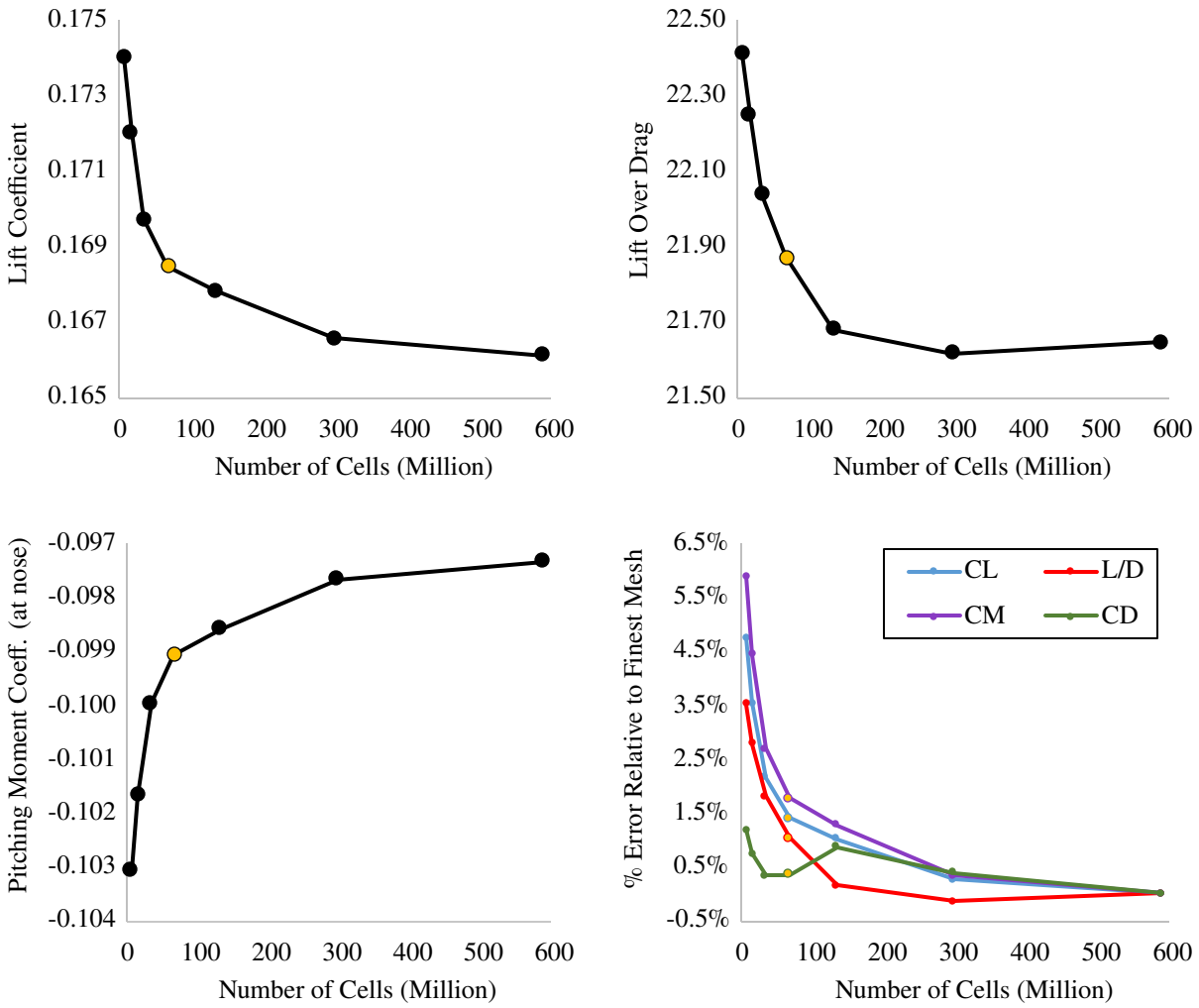


Fig. 7 Sensitivity of key responses to STAR-CCM+ grid refinement with selected grid shown using a yellow marker

and the change in C_L is less than 0.001 over 1000 iterations. Both conditions needed to be satisfied for stopping.

We used an unstructured polyhedral grid with prism layers to capture the boundary layer gradients over the surfaces. Through simple turbulent flat plate relations, we were able to calculate the total boundary layer thickness, near wall spacing, and number of prism layers required to maintain a wall $y^+ < 1$ over a majority of the outer mold line. We conducted a grid refinement study for the baseline BWB geometry at $M_\infty = 0.74$, 42,000 ft, angle of attack at 2.75°. Figure 7 shows the trends for the main responses.

We observe that grid convergence was achieved around 300 million cells, however, it was impractical from a cost perspective to use those grid settings for the optimization effort and drag polar generation, which would have required $\mathcal{O}(100)$ CFD runs. As such, we selected a set of coarser grid settings that generated a 68 million cell mesh for the baseline (shown with a yellow marker in Fig. 7). The computational cost of these grid settings was about 8 times lower than that for the 300 million cell grid and about 16 times lower than that for the 600 million cell grid. Additionally, the

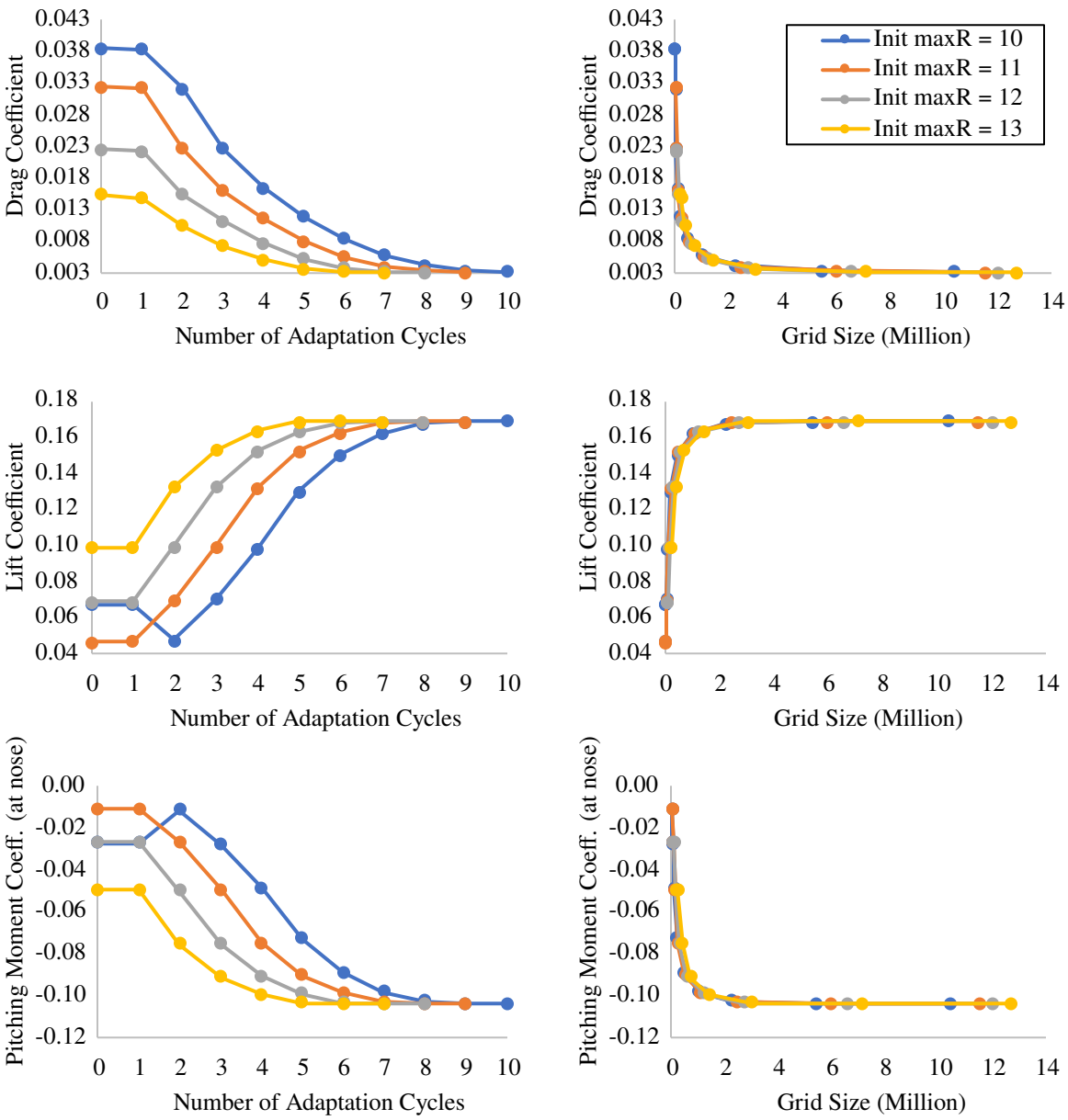
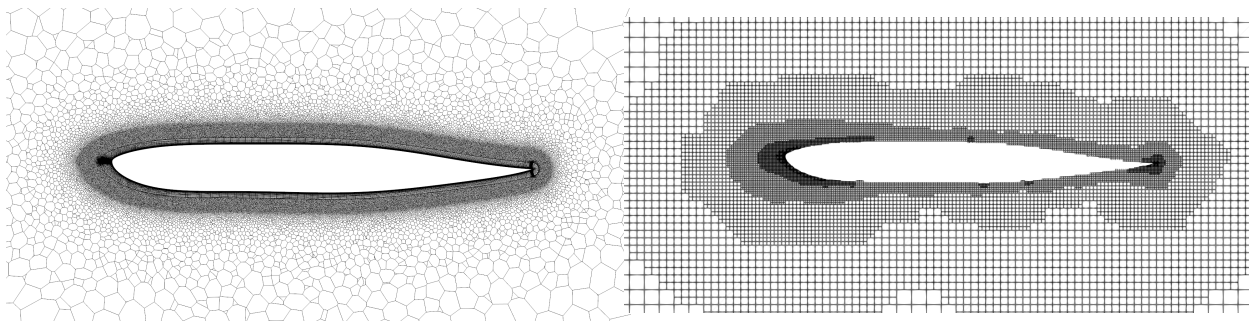


Fig. 8 Sensitivity of key responses to Cart3D grid refinement

error in L/D for the selected grid, relative to the result from the finest grid, was only 1%, with a 1.8% error in C_M. As such, this pragmatic choice allowed for acceptable levels of accuracy and computational cost.

A snapshot of the grids for the RANS and Euler cases is shown in Fig. 9. Note that for both the Euler and RANS cases, only the clean airframe without the fan array geometry was modeled. For the inviscid CFD cases, we used NASA's Cart3D[†]. We selected the Barth-Jespersen [42] limiter and the Van-Leer flux function [43], with the CFL number limited to 1. We leveraged Cart3D's adaptive mesh refinement capability and specified a stopping condition of 350 iterations for each cycle of the flow solver and 200 iterations for each cycle of the adjoint solver. Similar to the RANS cases, we conducted a grid sensitivity study at $M_\infty = 0.74$ and angle of attack 2.75°, where we varied the initial grid coarseness through the "maxR" parameter and adjusted the number of adaptation cycles. Based on the convergence of C_D, C_L and C_M shown in Fig. 8, we selected an initial grid refinement level of 13 with 6 adaptation cycles, yielding a final grid size of roughly 7 million cells.



(a) STAR-CCM+ volume grid (RANS) **(b) Cart3D volume grid (inviscid)**

Fig. 9 Images showing a slice of the STAR-CCM+ and Cart3D volume grids at the $y = 2.0$ m spanwise station for the selected mesh settings

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IV. Results and Discussion

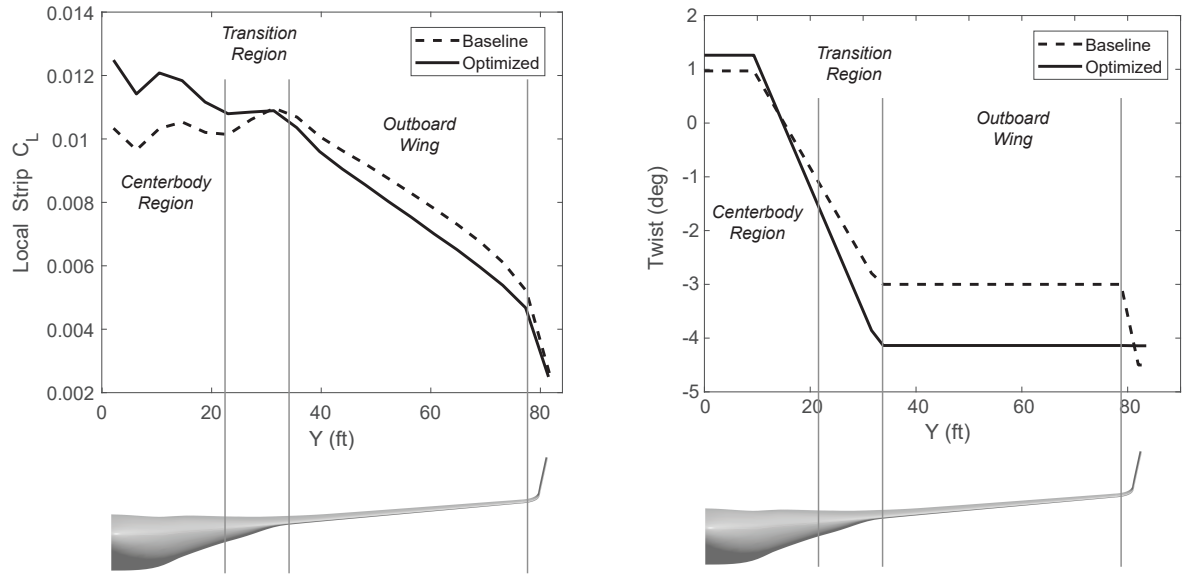
A. Optimized Geometry Characteristics

Table 5 presents a comparison of the baseline and constrained optimized configuration aerodynamic performance characteristics. The weightings we used on the objective function for L/D and $C_{M_{cg}}$ appear to bias the optimizer towards the latter response. As such, we see that the optimized design is able to achieve a negligible pitching moment at the center of gravity with a minimal change in L/D, while maintaining the target C_L to an acceptable degree of tolerance. To reduce $C_{M_{cg}}$, the optimizer reduced the sectional lift generated by the outboard wing, as seen in Fig. 10a by reducing the twist of the outboard wing relative to the baseline, as shown in Fig. 10b. The moment arm from the center of lift to

[†]https://www.nas.nasa.gov/software/cart3d.html

Table 5 Baseline versus Optimized Performance at $M_\infty = 0.74, 42,000$ ft

Parameter	Baseline	Optimized
C_L	0.175	0.176
C_D	0.0078	0.0079
L/D	22.3	22.3
$C_{M_{cg}}$	-7.8×10^{-3}	-3.2×10^{-5}
Angle of attack	2.78°	3.43°
Static margin	5.7%	5.4%
Objective function	38.9	-22.3



(a) Spanwise lift distribution

(b) Spanwise twist distribution

Fig. 10 Comparison between baseline and optimized lift and twist distributions

the center of gravity is larger for the outboard wing relative to the centerbody. Given the C_L constraint, the loss in lift from the outboard wing had to be compensated by an increase in lift from the centerbody, which is why the centerbody section features a higher twist in the optimized design with a higher sectional lift as seen in Fig. 10b and Fig. 10a. The airfoil shapes are virtually unchanged between the baseline and optimized configurations and this comparison is not shown for brevity. One of the likely factors for this behavior is the lack of wave drag given the low freestream Mach number, which means frictional and induced drag effects dominated. That being said, given the fixed planform area, fixed C_L, and the chosen design variables, there was likely not much room for improvement in drag relative to the well-performing baseline. Coupled with this factor is the stronger weighting we assumed for the pitching moment in the objective function, which, compared to L/D, had a higher potential for change compared to the baseline.

Figure 11 presents a snapshot of the multi-fidelity CFD drag polars for the optimized configuration. Although the drag polars were generated for multiple Mach-altitude combinations (varying Mach and Reynolds number), the polars presented in Fig. 11 are for specific Mach altitude combinations that have roughly the same Reynolds number (at cruise conditions).

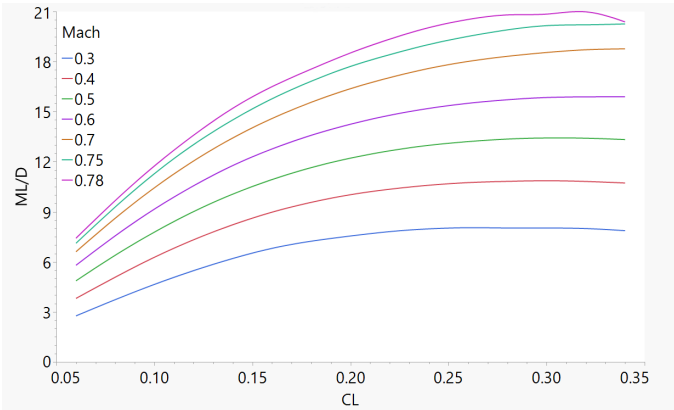


Fig. 11 Snapshot of the multi-fidelity polars showing $M_\infty L/D$ curves at cruise Reynolds number

B. System Level Performance Characteristics

Table 6 provides a high level summary of the system level performance characteristics of the IZEA concept for the design and economic missions, including their respective reserve missions. The detailed weight breakdown of the aircraft is shown in Table 7. One important observation to note is that the cruise C_L of the vehicle is lower than the C_L for maximum L/D seen in Fig. 11. As such, the vehicle is not cruising at peak aerodynamic efficiency, which is an area for further design effort in the future. Secondly, the final cruise C_L is very close to initial estimate used for the optimization target C_L. As such, the trim drag at cruise is still expected to be small given the effectively zero pitching moment at the center of gravity for the design conditions. The design mission range is 2,206 nmi, which is over 200 nmi above the target range of 2,000 nmi. Additionally, there is a small margin of approximately 69 lbs in usable fuel available when comparing the main + reserve mission total fuel to the assumed limit of 6,390 lb. As mentioned above, an empty weight margin is also included in the system level model, which is called out separately from the propulsion system group in Table 7, unlike in Table 4, where the margin was baked into the final propulsion and system weight estimate. These margins in range, fuel, and empty weight provide a buffer against potential changes in propulsion system weights, available fuel limits, aerodynamic performance capability, etc., as the design is refined further over the years and the sub-system level studies mature over time to yield revised performance numbers. As such, these initial performance estimates are quite encouraging.

To provide some context to the IZEA numbers, we compared this vehicle with our in house notional model of the Embraer E190-E2 regional jet. Key differences between the two vehicles include the type of fuel used, the payload, and design/maximum range. These differences make an apples-to-apples comparison of just fuel burn rather difficult. As such, to normalize these differences and bring the two aircraft to a more level playing field, we define the energy burn metric as follows:

$$\text{Energy Burn Metric} = \frac{\text{LHV} \times \text{Block Fuel}}{\text{Payload}} \cdot \frac{1000 \text{ nmi}}{\text{Range (nmi)}} \tag{4}$$

This metric represents the energy required to carry one pound of payload for 1,000 nmi. If the Lower Heating Value (LHV) has units of BTU/lb, then the units of the energy burn metric are BTUs per pound of payload. This metric allows us to compare the energy efficiency of both aircraft under normalized conditions, instead of directly comparing fuel burn, which would naturally favour the IZEA concept given the lower density of LH_2 fuel compared to Jet-A.

Table 8 compares the IZEA BWB specs to the notional E190-E2 model, where the latter was run at an off-design range of 2,206 nmi to match the IZEA BWB design range. Note that the aircraft is capable of flying much further and can carry more fuel than what is needed for this off-design case. The IZEA concept has a roughly 34% lower energy burn metric compared to the notional E190, although it is important to emphasize that we are comparing a 2050 EIS concept with a 2018 EIS aircraft, so this energy efficiency difference is quite high. Nevertheless, this difference is in the expected ballpark for such a comparison and showcases the high potential of the integrated technologies in allowing us to meet ambitious fuel and emissions targets in the future.

Table 6 System level performance for the IZEA concept (design and economic: main and reserve missions)

Metric	Design Main | Reserve	Econ. Main | Reserve
Range (nmi)	2,206 | 200	900 | 200
Cruise Mach	0.74 | 0.60	0.74 | 0.60
Start of Cruise Altitude (ft)	41,108 | 20,000	41,568 | 20,000
End of Cruise Altitude (ft)	41,732 | 20,000	41,753 | 20,000
Max Cruise L/D	20.8 (C_L = 0.172) | N.A.	20.7 (C_L = 0.171) | N.A.
Payload (lb)	25,200	25,200
Fuel Burn (lb)	4,946 | 1,375	2,169 | 1,236
Operating Empty Weight (lb)	88,273	88,273
Ramp Weight (lb)	119,795	116,879

Table 7 Detailed weight breakdown of the IZEA concept

Component	Weight (lb)
Outboard Wing (incl. transition)	12,692
Winglet	628
Centerbody	19,700
Landing Gear	4,662
Structures Total	*37,682*
Engine Components	20,573
Fuel System Tanks and Plumbing	4,260
Propulsion Total	*24,833*
Surface Controls	2,097
APU	0
Instruments	694
Hydraulics	1,771
Electrical	2,098
Avionics	2,078
Furnishings and Equipment	6,193
Air Conditioning	1,541
Anti-Icing	121
Systems and Equipment Total	*16,595*
Crew and Baggage	915
Unusable Fuel	575
Engine Oil	96
Passenger Service	1,846
Cargo Containers	875
Additional Operating Items Total	*4,307*
Payload Total (112 pax @225 lb/pax)	*25,200*
Design Mission Fuel	4,946
Reserve Mission Fuel	1,375
Fuel Total	*6,321*
Empty Weight Margin	*4,857*
Ramp Weight	*119,795*

Table 8 Comparison of the IZEA concept to current state-of-the-art regional jet

	IZEA BWB	Notional E190-E2
Range (nmi)	2,206	2,206
Payload (lb)	25,200	21,780
Ramp Weight (lb)	119,795	115,556
Block Fuel (lb)	4,946	18,167
LHV (BTU/lb)	51,580	18,580
Energy Burn Metric (BTU/lb)	4,589	7,025

V. Conclusions

This paper introduced the IZEA aircraft concept, a hybrid fuel cell + turboelectric system architecture integrated into a liquid hydrogen-fueled blended wing body. The IZEA Light was introduced as the smallest fuel cell variant of this aircraft and modeled at the system level to define high-level requirements for airplane propulsion, weights, and aerodynamics. The modeling methods for the aircraft system, weights, and propulsion were described in brief detail. A detailed configuration layout and CAD geometry were developed, and an aerodynamic optimization method was derived and conducted to improve drag performance and trim. The requirements and methods derived here will be used to conduct further research into this type of aircraft design, including detailed electrical distribution and thermal system architecting, fuel cell and power system modeling, hydrogen tank designs, and motor/inverter drive designs specific to the requirements of this airplane. As component designs are completed, and models created, an integrated system-level optimization model will be created for 3 different variant concepts within the hybridization design space with the goal of outlining the potential trade-offs between achieving zero emissions and minimizing total aircraft energy burn.

Acknowledgments

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